Cantera vs NN | Test IC 69 | NRMSE (util.py)

T0 = 935.0 K P0 = 2.045 atm phi0 = 0.523 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	9.88e-07	1.28e-06	1	185	1.53e-06	1	1	0.967	0.997	0	Complete
2 P-only DeepONet10	**10k**	2.83e-03	8.45e-03	9.87e+06	5.45e+07	6.11e-03	7.94e+07	1.15e+04	139	52.9	0	Complete
2 P-only DeepONet10	**long**	9.69e-07	1.29e-06	0.999	0.872	1.56e-06	0.979	0.997	0.994	1	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	9.70e-07	1.29e-06	1	0.492	1.55e-06	0.998	0.999	0.988	0.999	0	Complete
3 ANN10 + P-only DeepONet10	**long**	9.69e-07	1.29e-06	1	0.796	1.56e-06	0.995	0.997	0.992	1	0	Complete
4 T/P DeepONet10	**10k**	9.45e-07	1.27e-06	1	2.28	1.55e-06	0.952	0.963	0.997	1	0	Complete
4 T/P DeepONet10	**long**	3.65e-06	5.13e-06	0.853	1.79	6.38e-06	1.04	3.39	3.99	4.75	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	3.86e-04	5.04e-04	1	13	5.70e-04	0.999	8.88	94.2	1.33e+03	0	Complete
5 ANN10 + T/P DeepONet10	**long**	3.81e-05	4.88e-05	0.575	20.2	5.71e-05	7.21	31.3	29.7	57.5	0	Complete
6 T/P/ϕ DeepONet10	**10k**	2.78e-05	2.32e-05	87.3	109	1.84e-05	399	35.7	7.04	8.79	0	Complete
6 T/P/ϕ DeepONet10	**long**	9.71e-07	1.29e-06	0.505	0.95	1.56e-06	0.868	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	9.40e-06	8.94e-06	18.9	16.9	8.37e-06	52.1	11.3	4.03	5.05	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	1.03e-06	1.25e-06	156	0.854	1.56e-06	3.06	0.983	0.994	1	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.